

Requirement Specification Document

E-Commerce Order Management Database System

Week 1 – Requirement Analysis and Customer Database Module

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Project Title	Real-Time Hospital Management System
Week	Week 1

1. Introduction

1.1 Purpose

The purpose of this document is to analyze and specify the business requirements of the Real-Time Hospital Management System. The system is designed to manage patient records, doctor details, appointments, treatments, and billing efficiently

1.2 Project Scope

The system allows hospitals to maintain patient information, schedule appointments, assign doctors, record treatments, and generate bills. It provides a centralized database for efficient hospital management..

1.3 Intended Users

- Patients – Book appointments and view medical records.
- Doctors – View patient details and update treatments.
- Receptionists – Register patients and schedule appointments.

2. Business Requirements

The system should:

- Maintain patient records.
- Manage doctor information.
- Schedule appointments.
- Store treatment history.
- Generate patient bills.

3. Functional Requirements

Functional requirements describe what the system must do.

ID	Requirement Description
FR-1	The system shall allow patient registration.
FR-2	The system shall allow doctors to update patient diagnoses.
FR-3	The system shall schedule appointments.
FR-4	The system shall generate bills automatically.
FR-5	The system shall maintain treatment history.
FR-6	The system shall generate hospital reports.
FR-7	The system shall allow searching patient records.
FR-8	The system shall maintain doctor schedules.

4. Non-Functional Requirements

- **Performance:** Patient records should be retrieved within 2 seconds.
- **Security:** Patient data must be encrypted and secure.
- **Reliability:** The system should be available 24×7.
- **Usability:** The interface should be simple and user-friendly.
- **Scalability:** The database should support thousands of patients.

5. Customer Database Module – Data Requirements

Since this week's focus includes the Customer Database Module, the following entity and attributes have been identified:

Attribute	Data Type	Description
patient_id	INT (Primary Key)	Unique Patient ID
patient_name	VARCHAR(100)	Patient Name
age	INT	Patient Age
gender	VARCHAR(10)	Gender
address	VARCHAR(255)	Residential Address
registration_date	DATE	Registration Data

6. Assumptions and Constraints

- The system will use MySQL or PostgreSQL.
- Each patient has a unique Patient ID.
- Only authorized staff can access patient records.
- This document covers only the Patient Database Module for Week 1.

7. Conclusion

This document defines the initial requirements for the **Real-Time Hospital Management System**, focusing on the **Patient Database Module**. It provides the foundation for designing the ER diagram, database schema, and implementation in future weeks.